

Algebra 1
Unit 8
Lesson 1 Practice

Name _____
Date _____
Period _____

1. The equations $y = -3(x + 2)(x - 3)$ and $y = -3\left(x - \frac{1}{2}\right)^2 + \frac{75}{4}$ both represent the same quadratic. Rewrite both equations in standard form, $y = ax^2 + bx + c$. Show your work.

$y = -3(x + 2)(x - 3)$	$y = -3\left(x - \frac{1}{2}\right)^2 + \frac{75}{4}$
$y = -3(x^2 - 3x + 2x - 6)$ $y = -3(x^2 - x - 6)$ $y = -3x^2 + 3x + 18$	$y = -3\left(x - \frac{1}{2}\right)\left(x - \frac{1}{2}\right) + \frac{75}{4}$ $y = -3\left(x^2 - x + \frac{1}{4}\right) + \frac{75}{4}$ $y = -3x^2 + 3x - \frac{3}{4} + \frac{75}{4}$ $y = -3x^2 + 3x + 18$

2. The equations $y = -(x + 6)(x - 3)$ and $y = -(x - (-6))(x - 3)$ are equivalent. How can both x -intercepts be determined if the equation $y = -(x + 6)(x - 3)$ is given?

The x -intercepts can be determined by using Zero Product Property. Set each factor equal to zero and then solve to find the x -intercepts.

3. The equation $y = -2x^2 - 4x + 30$ can be used to determine the y -intercept. Show how each of the three forms can be used to find the y -intercept.

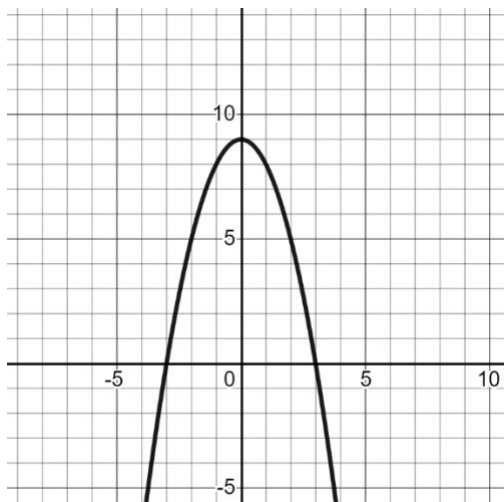
$y = -2x^2 - 4x + 30$	$y = -2(x + 5)(x - 3)$	$y = -2(x + 1)^2 + 32$
$y = -2(0)^2 - 4(0) + 30$ $y = -2(0) - 4(0) + 30$ $y = 30$	$y = -2(0 + 5)(0 - 3)$ $y = -2(5)(-3)$ $y = 30$	$y = -2(0 + 1)^2 + 32$ $y = -2(1)^2 + 32$ $y = -2 + 32$ $y = 30$

4. A table of values for $y = \frac{1}{4}(x - 2)^2 + 8$ is shown. What features can be found using the table of values?

- ☐ x -intercepts
☒ y -intercepts
☒ vertex
☒ axis of symmetry
☒ increasing/decreasing
☐ positive/negative

x	y
-3	14.25
-2	12
-1	10.25
0	9
1	8.25
2	8
3	8.25

5. The graph of a quadratic equation is shown. Use words, ordered pairs, or symbols to describe the location of each key feature.



Key Feature	Location of Key Feature
x -intercepts	$(-3,0)$ $(3,0)$
y -intercepts	$(0,9)$
vertex	$(0,9)$
axis of symmetry	$x = 0$
increasing	$x < 0$
decreasing	$x > 0$

6. A table of values for $y = (x - 1)^2$ is shown. What features can be found using the table of values?

- ☒ x –intercepts
- ☒ y –intercepts
- ☒ vertex
- ☒ axis of symmetry
- ☒ increasing/decreasing
- ☒ positive/negative

x	y
-1	4
0	1
1	0
2	1
3	4

Use the graph of the quadratic equation for questions 7-10.

7. Describe where the parabola is increasing.
 $x > 4$
8. Describe where the parabola is decreasing.
 $x < 4$
9. Draw the axis of symmetry for the parabola on the graph.
10. Circle the vertex, x –intercepts, and y – intercepts of the parabola. Label each point as vertex, x –intercepts, and y – intercepts.

